

Individual Project — Final Report

Evaluation Form — 150 points

EEB 603 Reproducible Science — Boise State University

Fall 2026

Student: _____

Project type: _____

Date: _____

Evaluator: _____

1. Overview of the Assignment

This rubric is designed to evaluate individual reports in which students develop and document a reproducible workflow tailored to their thesis project or a selected publication. The rubric accommodates both **full projects** (with real data and completed analyses) and **pre-studies** (for students who have not yet generated data but are preparing for thesis research).

The report is the second of the three stages of the individual project. The first stage — the **project proposal**, worth 50 points — is evaluated on its own form, and the third — the **oral presentation**, worth 100 points — on another; both scores are carried forward into the summary at the end of this document. Sections 1–3 below are weighted more lightly than the execution sections precisely because the framing, data identification and workflow planning were already assessed at the proposal stage; what is assessed here is what the student actually built.

2. Rubric

1. Framing of the Scientific Problem — 15 points

Evaluates clarity of scientific context and purpose.

Criteria

- **Problem statement and rationale (5 pts):** Clear introduction to the scientific question, its relevance, and its context within the field.
- **Objectives and hypotheses (5 pts):** Well-defined objectives, hypotheses, or guiding research questions.
- **Integration into the scientific process (5 pts):** Demonstrates understanding of how the project or pre-study fits within the broader research workflow.

Comments:

Score: _____ / 15

2. Use and Identification of Data Sources — 15 points

Accommodates both existing data (for projects) and conceptual data exploration (for pre-studies).

Criteria

- **Published/existing data or literature basis (6 pts):**
 - *Projects*: Identification of relevant published datasets and where they are deposited.
 - *Pre-studies*: Identification of key literature describing the data landscape and availability.
- **Preliminary, simulated, or intended data (3 pts):** Description of thesis-generated, preliminary, or hypothetical data including characteristics and constraints.
- **Assessment of reproducibility of existing work (6 pts):**
 - *Projects*: Attempt or assessment of reproducing published analyses.
 - *Pre-studies*: Critical evaluation of potential reproducibility based on data availability and published methods.

Comments:

Score: _____ / 15

3. Data Management Workflow — 25 points

Evaluates planning and documentation across the data life cycle.

Criteria

- **Coverage of the data life cycle (Create, Process, Document, Preserve, Share, Reuse) (12 pts):**
 - *Projects*: Concrete workflow applied to actual data.
 - *Pre-studies*: Conceptual workflow outlining intended procedures once data become available.
- **Documentation practices (9 pts):** Clear description of metadata, file structure, naming conventions, versioning, and storage.
- **Field-specific standards (4 pts):** Appropriate reference to established standards (e.g., Donoho 2010; Smith et al. 2016).

Comments:

Score: _____ / 25

4. Reproducible Code & Computational Workflow — 45 points

This section accommodates variability in project maturity and is flexible for pre-studies.

Expectations based on project type

- **Full project:** Complete reproducible pipeline (import → tidy → transform → visualize → model → communicate).
- **Pre-study:** Demonstration of reproducible coding practices using example, published, or simulated data; partial workflows are acceptable if well-justified.

Criteria

- **Reproducible scripting structure (10 pts):** Use of RMarkdown, clear code chunks, relative paths, and good computational hygiene.
- **Demonstration of key data-analytic steps (15 pts):**
 - *Projects*: Full pipeline from raw data to outputs.
 - *Pre-studies*: Partial or prototype workflow using example data.
- **Use of reproducible tools and best practices (10 pts):** Includes version control, data integrity checks, automation, or workflow diagrams.
- **Communication and integration in RMarkdown (10 pts):** Clear presentation of results (real or example), integrated narrative, and appropriate figure/table inclusion.

Comments:

Score: _____ / 45

5. Reproducibility, Transparency, and Replicability — 30 points

Evaluates how well the student demonstrates or plans for reproducibility.

Criteria

- **End-to-end reproducibility (13 pts):** For projects: full execution from raw to final results. For pre-studies: clear plan or demonstration using example workflows.
- **Transparency and methodological justification (9 pts):** Clear explanation of why methods were chosen, supported by citations.
- **Effective use of RMarkdown/knitr or a well-structured document (in e.g. docx) (8 pts):** Integration of code, output, and narrative in a reproducible manner.

Comments:

Score: _____ / 30

6. Citations and Reference Use — 12 points

Criteria

- **In-text citations (6 pts):** All referenced methods, data sources, and concepts are cited in the text, not just listed.
- **Relevance and quality of references (3 pts):** References support methodological choices and enhance transparency.
- **Bibliography formatting (3 pts):** Consistent, complete reference list.

Comments:

Score: _____ / 12

7. Writing Quality & Organization — 8 points

Criteria

- **Clarity and logical flow (5 pts):** The report is well-structured, coherent, and easy to follow.
- **Professional writing and formatting (3 pts):** Clear figures and tables, appropriate labeling, academic tone, and minimal errors.

Comments:

Score: _____ / 8

3. Score Summary

Section	Points	Score
1. Framing of the Scientific Problem	15	
2. Use and Identification of Data Sources	15	
3. Data Management Workflow	25	
4. Reproducible Code & Computational Workflow	45	
5. Reproducibility, Transparency, and Replicability	30	
6. Citations and Reference Use	12	
7. Writing Quality & Organization	8	
REPORT TOTAL	150	

Individual Project — Combined Total

Stage	Points	Score
Project proposal (<i>carried forward from the proposal evaluation form</i>)	50	
Final report (<i>from the table above</i>)	150	
Oral presentation (<i>carried forward from the presentation evaluation form</i>)	100	
INDIVIDUAL PROJECT TOTAL	300	

4. Feedback Summary

Strengths

Areas for Improvement

Progress Since the Proposal

How far did the student act on the feedback given at the proposal stage?

Final Comments